



University of Stuttgart

Institute for Control Engineering of Machine
Tools and Manufacturing Units (ISW)



The Future of Industrial AI

Model-Driven Digital Twins
and Autonomous Systems



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- Director of ISW, University of Stuttgart
- Director of **AI Software Engineering Academy** at Stuttgart University
- Board of **European Association for Programming Languages & Systems**
- Spokesperson for **AI-Powered Robotics** of Robotics Institute Germany
- Founder of **ACM/IEEE Engineering Digital Twins** conference
- Research interests
 - **Model-driven engineering**
 - **AI for engineering**
 - **Cyber-physical systems (CPSs)**
 - **Digital twins**





Head of Institute:

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Research fields

Software- and
Engineering Methods

Industrial Control
Engineering

Real-Time Communication
and Control Hardware

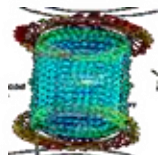
Drive Systems and
Motion Control

Mechatronic Systems
and Processes

Virtual Methods for
Production Engineering



Engineering
Methods

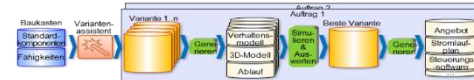


Simulation



Communication

Model-Based
Production Engineering



Product Influence



Production IT

Control

Algorithms
Operation
Architectures



Control
Engineering

Mechatronic Systems



Drive
Technology

Drive Control and Machine Technology



Digital Twins Yield Benefits of £850 Million Across a 10-Year Period¹

Manufacturing impact of DTs estimated² in the „low tens of billions of dollars“

- EU DT revenue forecast: **\$49.3B by 2030**²
- KR DT revenue forecast: **\$3.8B by 2034**³
- US DT-based savings of **\$27.2M / year**⁴
- BMW: **25% reduced time to plan factory ops**⁵
- BCG: **3% - 6% procurement cost reduction**⁶
- Capgemini: **16% better sustainability metrics**⁷
- Siemens: **30% service cost reduction, 8% output increase**⁸

Digital Twin Use Cases

30%

System
prediction

28%

System
simulation

24%

Asset
interoperability

21%

Maintenance

20%

System
visualization

9%

Product
simulation

1. <https://digitaltwinhub.co.uk/research-on-the-economic-benefits-of-digital-twins-for-integrated-transport-network-management/> (2024)

2. <https://www.marketsandmarkets.com/Market-Reports/europe-digital-twin-market-195973825.html> (2026)

3. <https://www.imarogroup.com/south-korea-digital-twin-market> (2025)

4. National Institute of Standards and Technology (NIST). Economics of Digital Twins. https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=958153 (2024)

5. TIME Magazine: How Digital Twins Are Transforming Manufacturing, Medicine and More (2021)

6. BGC: Using Digital Twins to Manage Complex Supply Chains (2024)

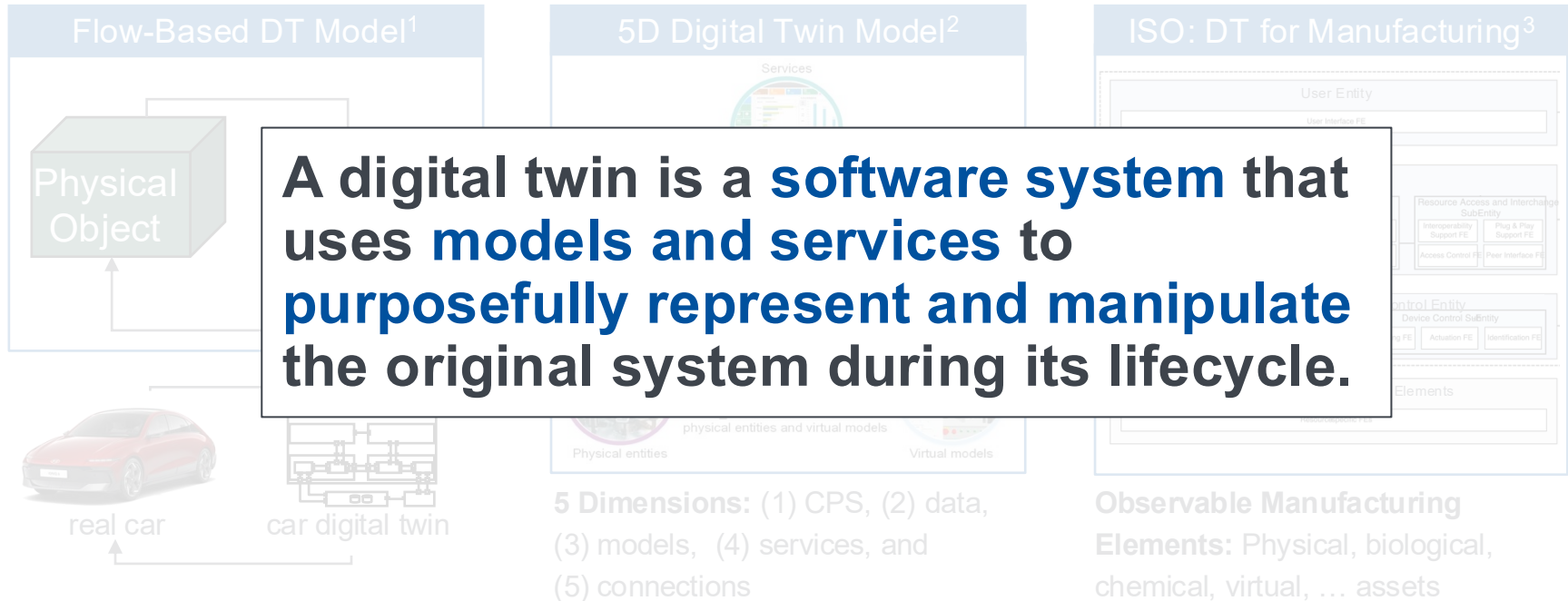
7. Capgemini: Digital Twins are a catalyst to fulfilling organizations' sustainability agenda (2022)

8. Siemens Digital Industries Software: Four pillars of the industrial IoT (2020)

Source: <https://www.jeffwinterinsights.com/>

Most Conceptualizations of Digital Twins are not too Useful

Combining flow-based definition with 5D model gives meaning to the concept



1. Kritzinger, W., Kamer, M., Traar, G., Henjes, J., & Sihn, W: Digital Twin in manufacturing: A categorical literature review and classification. IFAC-PapersOnLine, 2018.
2. Qi et al.: Enabling technologies and tools for digital twin. In: Journal of Manufacturing Systems, Elsevier, 2019
3. ISO 23247: Digital Twin Framework for Manufacturing, 2021.

Digital Twins (DTs) are Costly: Minimum \$ 1M / DT¹

Manufacturing consists of SMEs with limited software engineering expertise

LANDSCAPE²

- Manufacturing 27% of KR GDP
- **World-class companies:** Samsung, Hyundai, LG, ...
- **97.3%** of manufacturing companies **SMEs or smaller**
- Contribute 10% of GDP
- **Initiatives:** Digital Twin Korea, Digital Twin National Land, Smart Farms Testbed, POSCO, POSTECH PIAI & SMART, ...

CHALLENGES

- Capgemini: aero & defense invest **2.7% of AR** in DTs³
- Mid-complexity DTs (e.g., single production line): **\$500k - 2M**⁴
- Deployment: 3-6 mo.⁵, First value: +4 mo.⁶, Scale: +9 mo.⁵
- **Hampers adoption** and prevents more efficient use of systems
- Most manufacturing companies face **more urgent challenges**

LOW-CODE SOLUTIONS

- **Vertically integrated** model-driven engineering (MDE)
- **Tailored modeling techniques** for domain experts
- Often **little to no software engineering** required
- **Examples:** Appian, Google AppSheet, n8n, NodeRED, Mendix, Microsoft PowerApps, Oracle APEX, OutSystems

1. <https://www.ptc.com/en/blogs/corporate/roi-of-digital-twin-for-industrial-companies>

2. Ministry of SMEs and Startups. The Status of Small and Medium-Sized Enterprises (2026): <https://www.mss.go.kr/site/smba/office/ex/statDB/MainSubStat.do>

3. Capgemini. Aerospace and defense organizations show growing confidence in Digital Twin technology with 40% invest. increase. (2025)

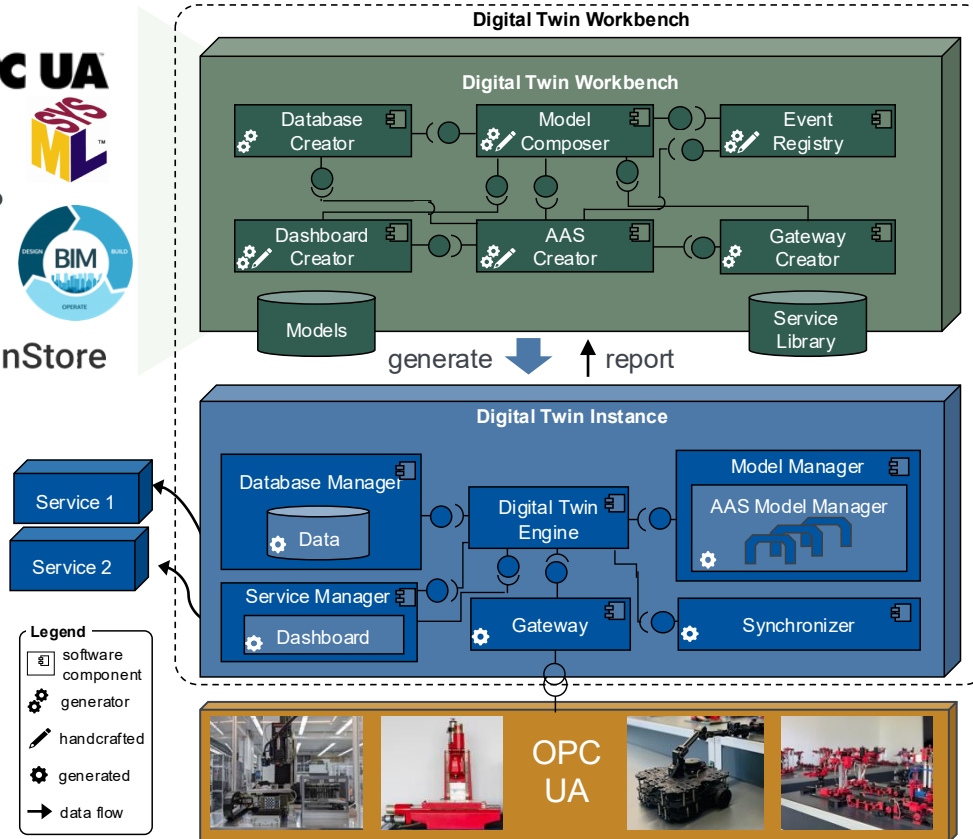
4. Medium. The Business Case for Digital Twin: Costs, Benefits, and ROI Explained. (2026)

5. Capgemini. REFLECTIOD Digital Twin For Asset Operators (2021)

6. Accenture. How digital twins enable autonomous operations (2022)

Our Workbench Employs Models as Blueprint and at Runtime

Ingests engineering models and services, produces digital twin instances



Digital Twin Workbench

- ingests **engineering models**
- instantiates DT **reference architecture**
- **creates** data infrastructure, UI
- generates **DT instance**

Digital Twin Implementation

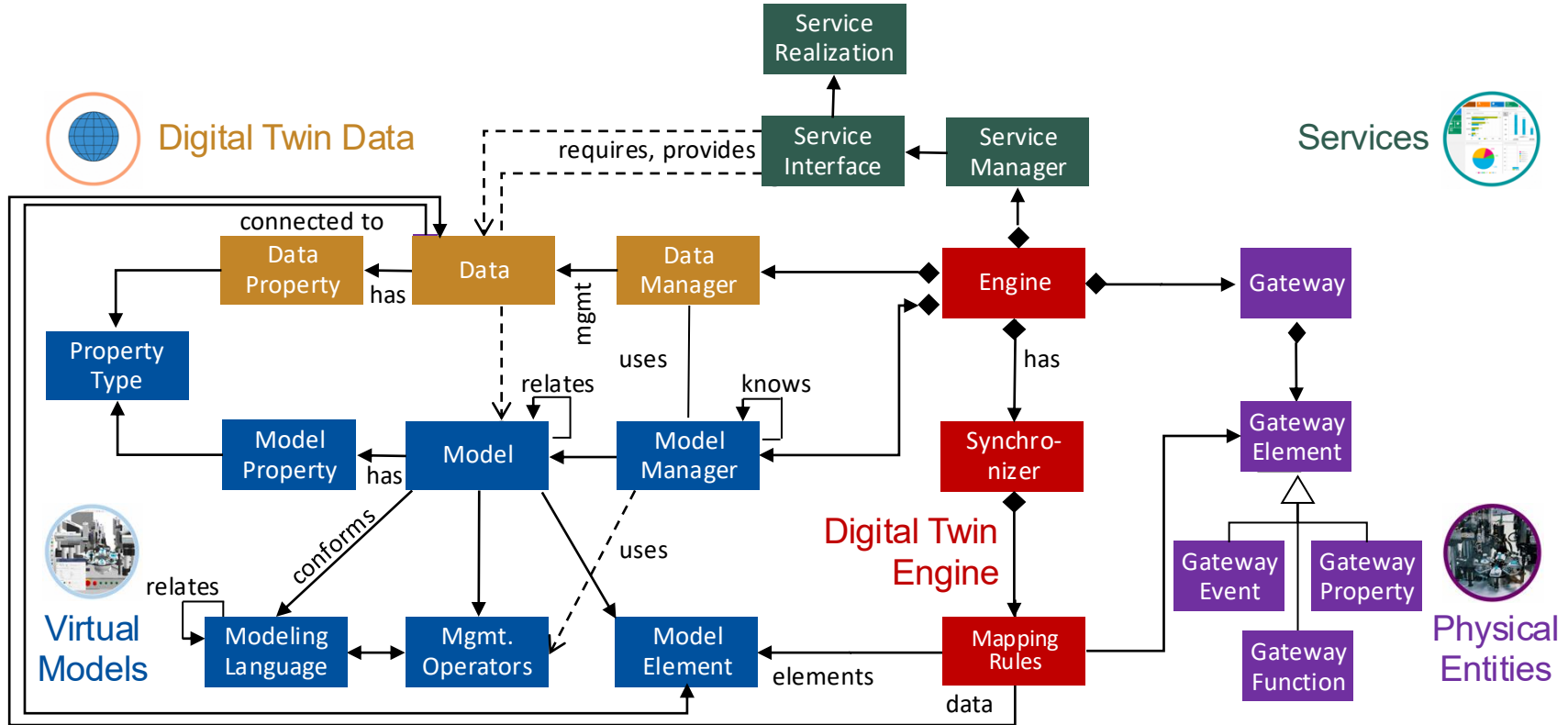
- **connects to CPS** via OPC UA
- ingests data & **controls**
- uses **added-value services**

Separation of Concerns

- **software engineers:** workbench, digital twin architecture, services
- **domain experts:** runtime models, configuration

A Reference Architecture for ISO 23247 Digital Twins in Manufacturing

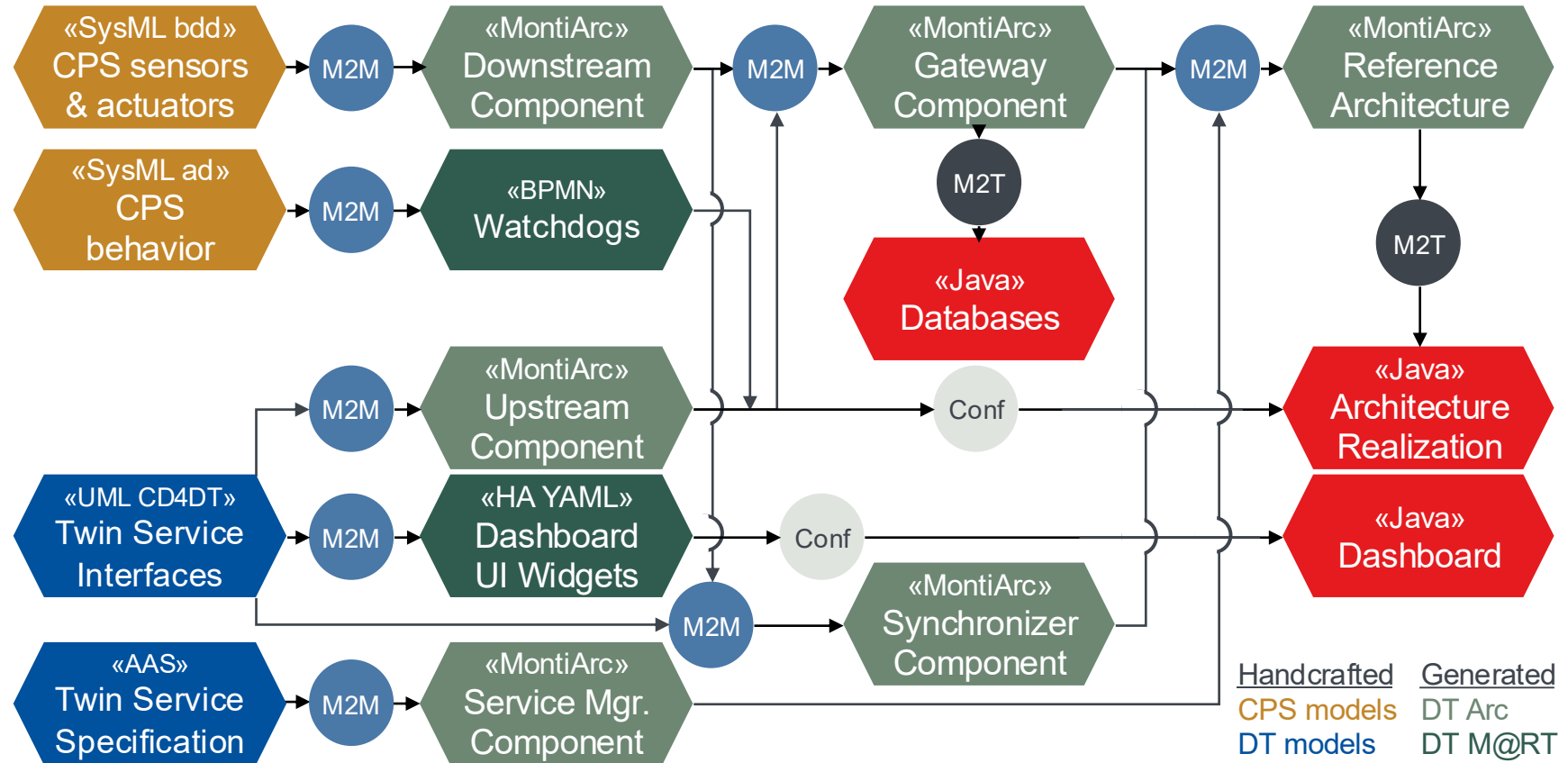
Based on the 5D model¹ and refined with modeling expertise



1. Pfeiffer, J., Zhang, J., Combemale, B., Michael, J., Rumpe, B., Wimmer, M., & Wortmann, A. (2025). Towards a unifying reference model for digital twins of cyber-physical systems. In 2025 IEEE 30th International Conference on Emerging Technologies and Factory Automation (ETFA). IEEE

The Digital Twin is Created via Model Transformations

A formalism transformation graph overview



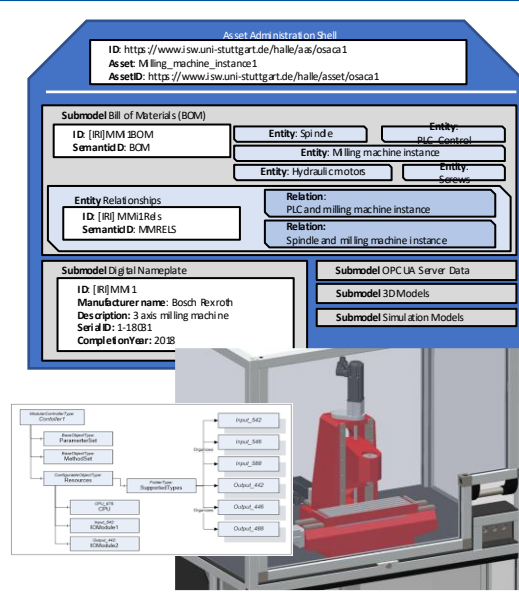
Case Study 1: OSACA 3-Axis Drilling Machine¹

Computing the CO₂ footprint of a drilling machine via AAS

PURPOSE

- Analyzing energy consumption vs OCL-defined thresholds to detect wear
- Manipulate machine state and spindle speeds to ensure maximum energy efficiency
- Models:** virtuos 3D model from virtual commissioning, OPC UA, Asset Administration Shell

MODELS IN THE DT



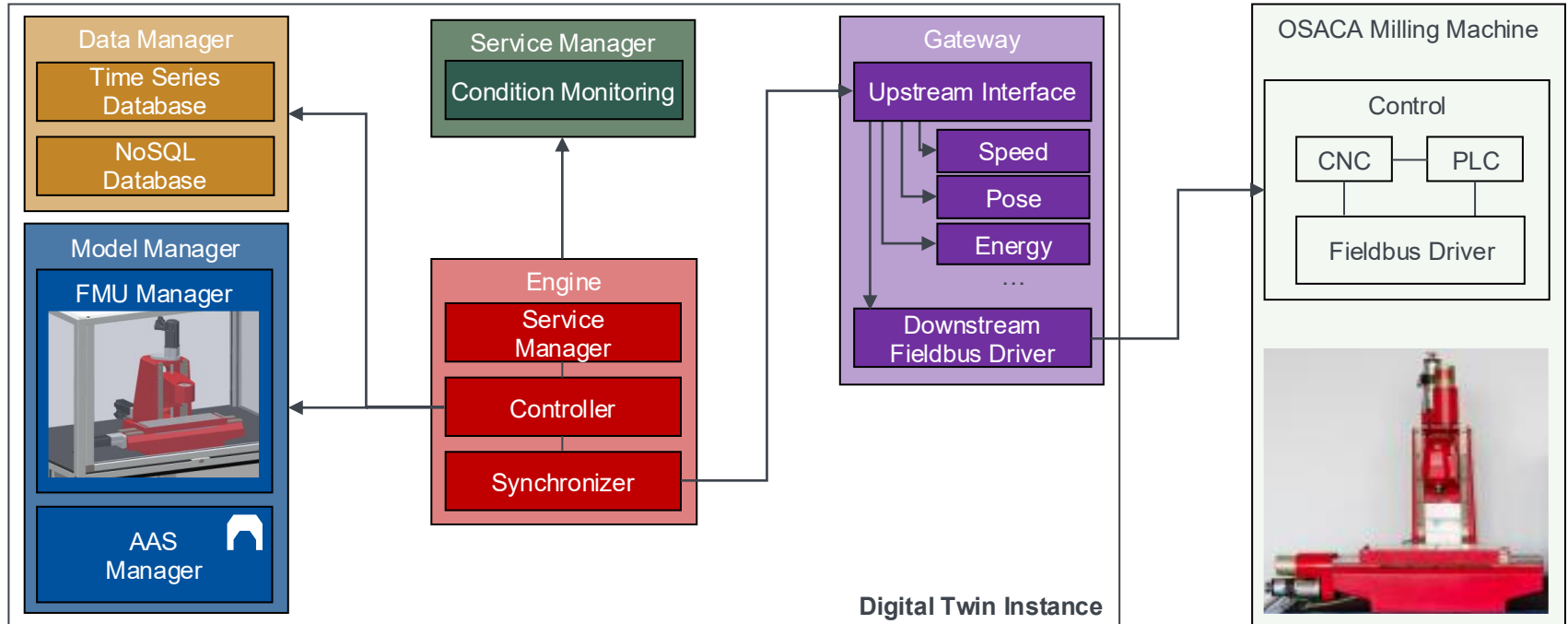
CYBER-PHYSICAL TWIN



1. Chen, S., Zhang, J., Jud, O., Klingel, L., Verl, A., Wortmann, A., & Ajdinović, S. (2026). A Software Architecture for Real-Time Digital Twins in Machining. In Proceedings of the 8th Workshop on Modeling and Simulation of Software-Intensive Systems.

Case Study 1: OSACA 3-Axis Drilling Machine¹

Incarnation of reference architecture for an OSACA sustainability use case



1. Chen, S., Zhang, J., Jud, O., Klingel, L., Verl, A., Wortmann, A., & Ajdinović, S. (2026). A Software Architecture for Real-Time Digital Twins in Machining. In Proceedings of the 8th Workshop on Modeling and Simulation of Software-Intensive Systems.

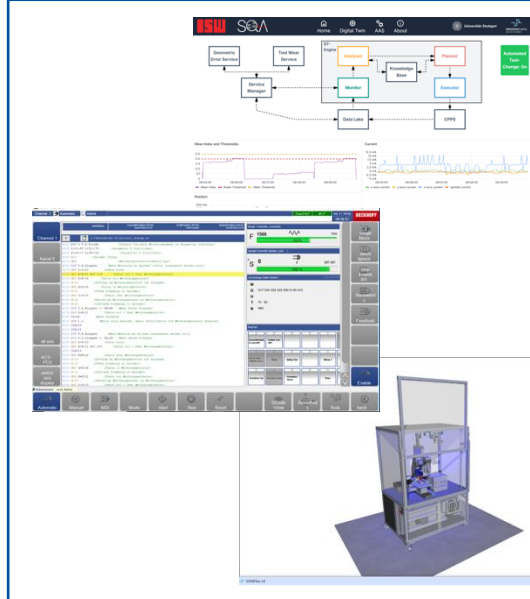
Case Study 2: FiveX 5-Axis Laser Cutting Machine¹

Enabling proactive hardware maintenance and quality control

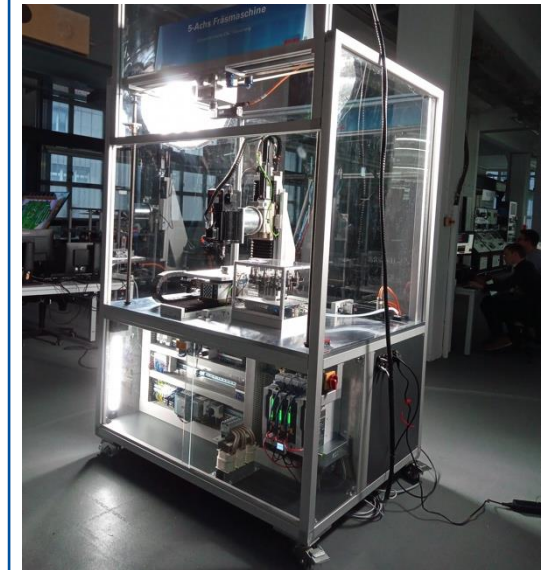
PURPOSE

- Automatically trigger the replacement of lenses
- Simulate heat distribution: predict if the material will warp or distort
- **Models:** Asset Administration Shell, OPC UA, SysML v2 block definition diagram, CAD, UML class diagram, machine skill models

MODELS IN THE DT



CYBER-PHYSICAL TWIN



1. Chen, S., Zhang, J., Jud, O., Klingel, L., Verl, A., Wortmann, A., & Ajdinović, S. (2026). A Software Architecture for Real-Time Digital Twins in Machining. In Proceedings of the 8th Workshop on Modeling and Simulation of Software-Intensive Systems.

Model-Driven Development of Digital Twins is more Efficient and Reliable

AI can help with this and digital twins can help AI

Introduction

Motivation

Digital Twins: Disambiguation

Low-Code Development of Digital Twins

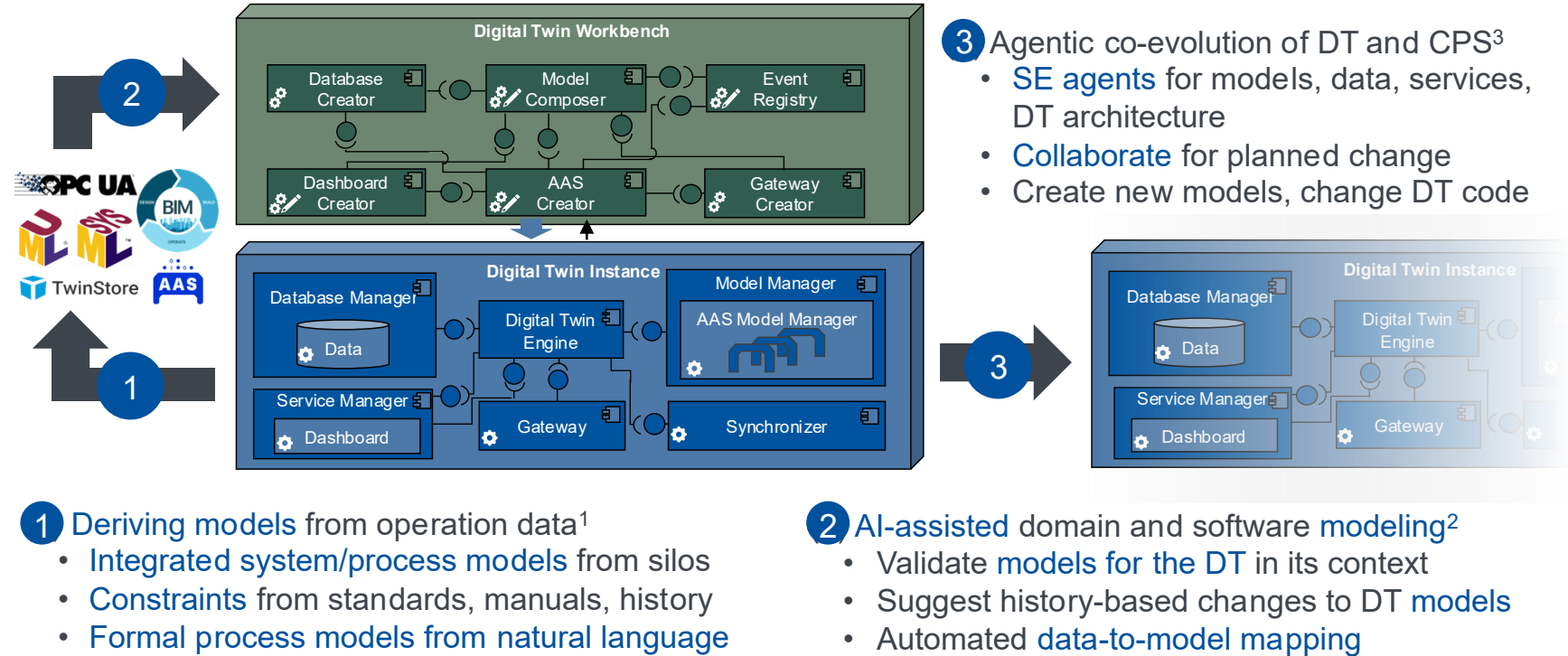
► **AI for Digital Twins**

Digital Twins for AI

Challenges

AI4DT: Break the 1M USD Per-Twin Cost Barrier That Hampers Adoption

By supporting the creating and integration of models and evolution of code

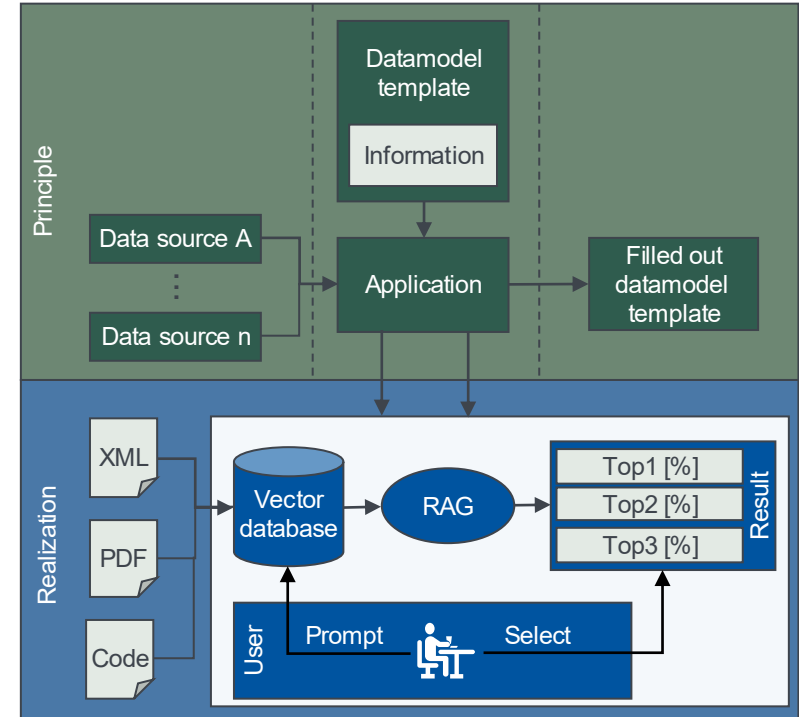


1. van der Aa, H., Di Ciccio, C., Leopold, H., & Reijers, H. A. (2019). Extracting declarative process models from natural language. In CAISE. Springer.
2. Kögel, S., Groner, R., & Tichy, M. (2016). Automatic Change Recommendation of Models and Meta Models Based on Change Histories. In ME@MODELS. ACM.
3. Dittler, D., Lierhammer, P., et al. (2022). An Agent-based Realisation for a continuous Model Adaption Approach in intelligent Digital Twins. arXiv:2212.03681.

AI4DT: AI-Powered Semantic Matching and Data Harmonization¹

Bridging interface mismatches between services and gateway

- Use embeddings of data interfaces to **semantically match different representations** of
 - gateway upstream vs downstream
 - downstream interface vs. OPC UA server
- **Threefold** approach
 1. **DBSCAN** cluster analysis
 2. **entity comparison**
 3. **LLM-assisted** evaluation
- **90% accuracy** on matching simple attributes
- But: many different data source kinds
- Integration of HiL learning with cold-start problem

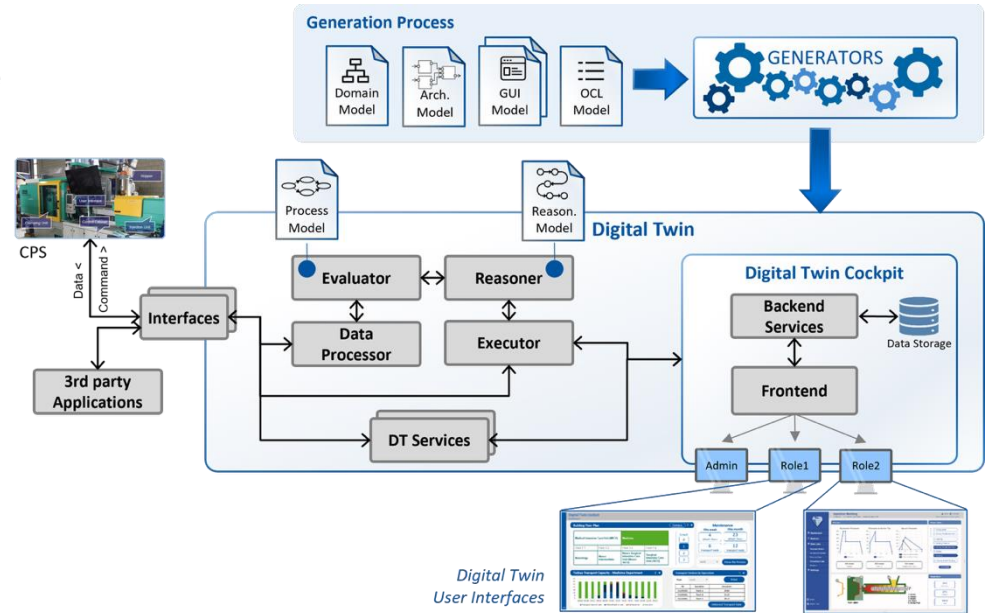
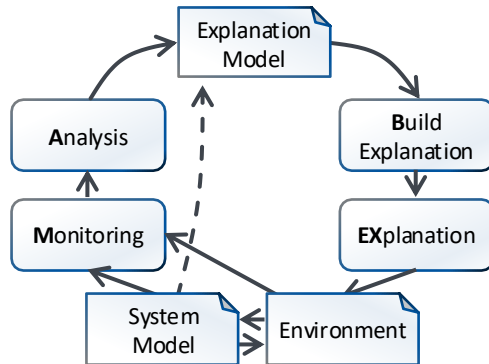


1. Ajdinović, S., Maisch, N., Kimmel, R., Rasouli, T., Lechler, A., Wortmann, A., & Riedel, O. (2025). AI-powered semantic matching and data harmonization for industrial applications with OPC UA & asset administration shells. In 16th International Conference on Mechanical and Intelligent Manufacturing Technologies (ICMIMT). IEEE.

AI4DT: Digital Twin Explains the Behavior of Its Cyber-Physical Twin¹

The operators of digital twins are domain experts who need domain explanations

- MAB-EX² makes systems **self-explainable**
- **Case-based reasoning**³ and models create explanations of system behavior
- Variant of the MAPE-K³ loop



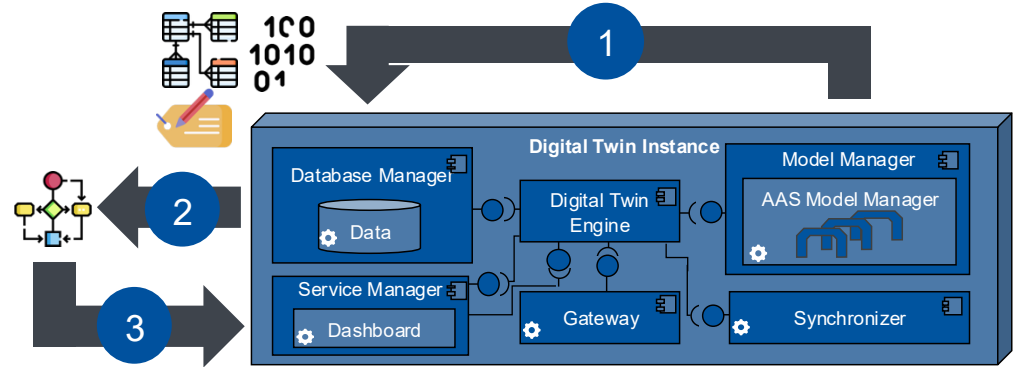
1. Michael, J., Schwammberger, M., & Wortmann, A. (2023). Explaining cyberphysical system behavior with digital twins. *IEEE Software*. IEEE
2. Blumreiter, M., Greenyer, J., Garcia, F. J. C., Kbs, V., Schwammberger, M., Sommer, C., ... & Wortmann, A. (2019). Towards self-explainable cyber-physical systems. In Proceedings of the 22nd International Conference on Model Driven Engineering Languages and Systems Companion.
3. Bolender, T., Bürvenich, G., Dalibor, M., Rumpe, B., & Wortmann, A. (2021). Self-adaptive manufacturing with digital twins. In: SEAMS. IEEE.

DT4AI: Digital Twins Can Improve and Facilitate Using AI in Manufacturing

By giving meaning to data, checking it and including human expertise

1 Data swamps to semantic data¹

- **Semantics for data** points from system models
- **Better labels** using CPS and process models
- Include **other data sources**: MES, ERP, simulation, ...



2 Continuous Assessment of Data Quality²

- **Automated plausibility checks using models** (ranges, expected signals, timing, ...)
- **Traceability**: metadata
- **Identify mismatches** between real and training data (**drift**)

3 Facilitate Continuous Learning³

- **Contextualize** new data
- **Validate new data** before retraining
- **Human in the loop**: ask operator in case of uncertainty, drift, or conflicting data

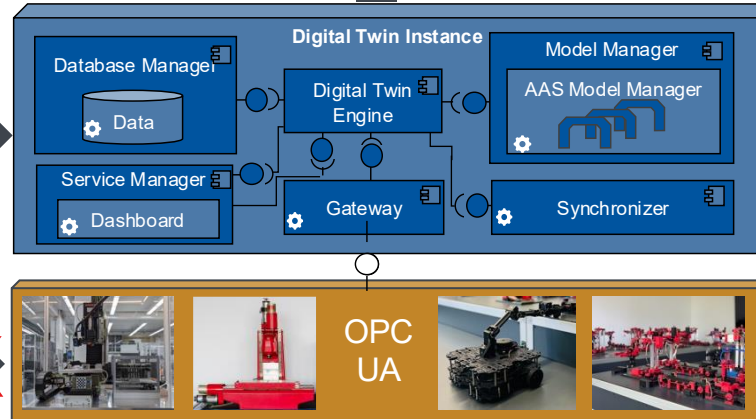
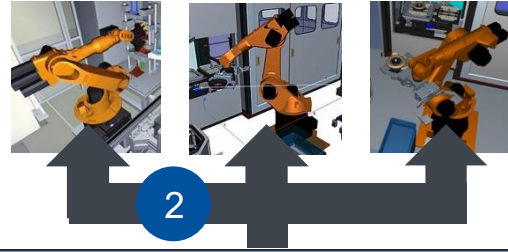
1. Becker, F., Bibow, P., Dalibor, M., Gannouni, A., Hahn, V., ... & Wortmann, A. (2021). A conceptual model for digital shadows in industry and its application. In International conference on conceptual modeling. Springer.
2. Saroj, A., Roy, S., Guin, A., & Hunter, M. (2025). Data Quality Assessment Process for Real-Time Data-Driven Traffic Microsimulation of Smart Corridor. In International Conference on Transportation and Development 2025.
3. Son, Y. H., Lee, E. S., & Son, J. Y. (2026). ACL-DT: adaptive continuous learning digital twin framework for resilient manufacturing. The International Journal of Advanced Manufacturing Technology, 1-14.

DT4AI: Digital Twins Make AI in Manufacturing More Reliable

By acting as smart safeguard for production systems

1 Digital twin as safeguard

- Layer between AI and shopfloor
- Validate AI decisions against explicitly modeled constraints
- Forecast machine health trajectory prior to AI execution



2 Exploration of alternative futures

- Automated parallel what-if sim
- Choose parameters based on machine and process history
- AI explain best course of action



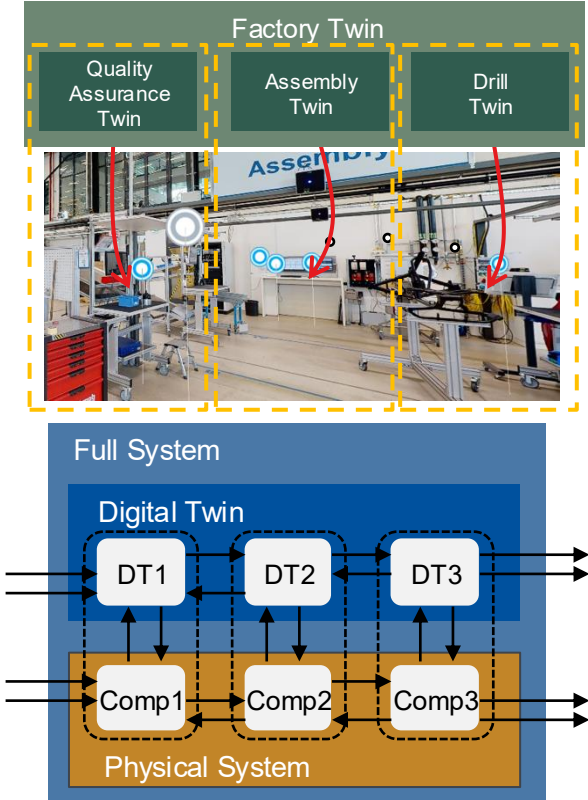
3 Analyze production reliability

- Simulate problems to connected systems (MES, ERP, SCADA)
- Evaluate effects of problems without having them
- Estimate effects of rare faults
- Learn causes and effects

Composition of Digital Twins is Essential

Combine simple digital twins to build more complicated digital twins

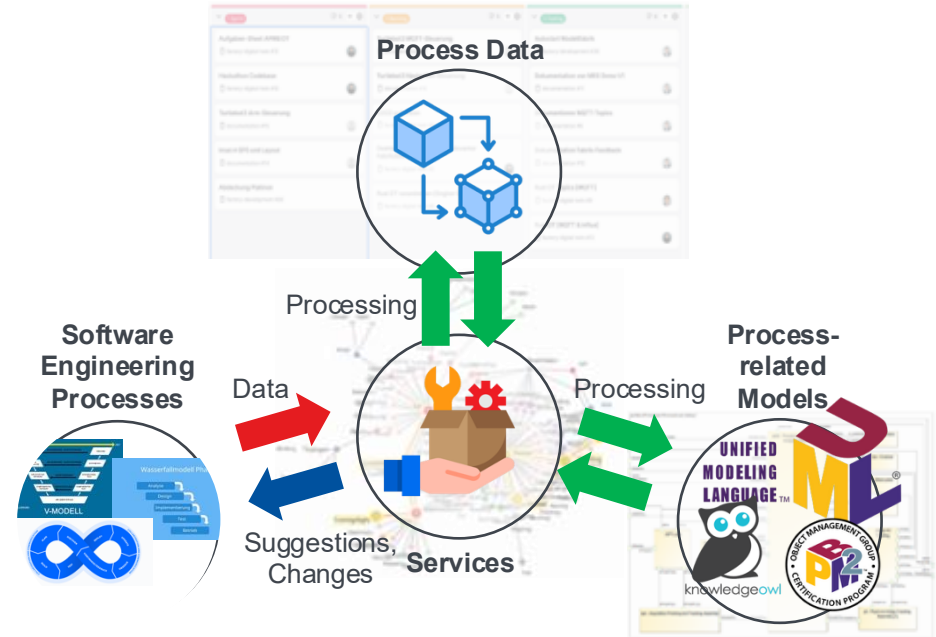
- **Composition vital** to many disciplines
- Examples: function composition (math), product composition (mechanics), software composition (CS), ...
- Efficient software engineering **without composition unthinkable**
- **Idea:** When physically composing sensor, machine tool, production line, and factory, compose their digital twin also
- Many open questions:
 - **How** to compose? (blackbox, whitebox, merge, federate, ...)
 - **Which** parts to compose? (models, data, components, ...)
 - **What** about conflicts? (factual, abstraction, control, ...)



Digital Twins for Software Engineering Processes¹

Towards real-time feedback to monitor and steer development processes

- Complex engineering processes slow technological innovation
- **Digital twin architecture for processes** similar to machine twins
 - process data, e.g., issue board, CI/CD
 - models, e.g., process, data, constraints
 - services, e.g., analysis, assignment, and change proposition
- **Autonomous AI-powered process support** activities through a standardized model
- SE agents interact with process twin

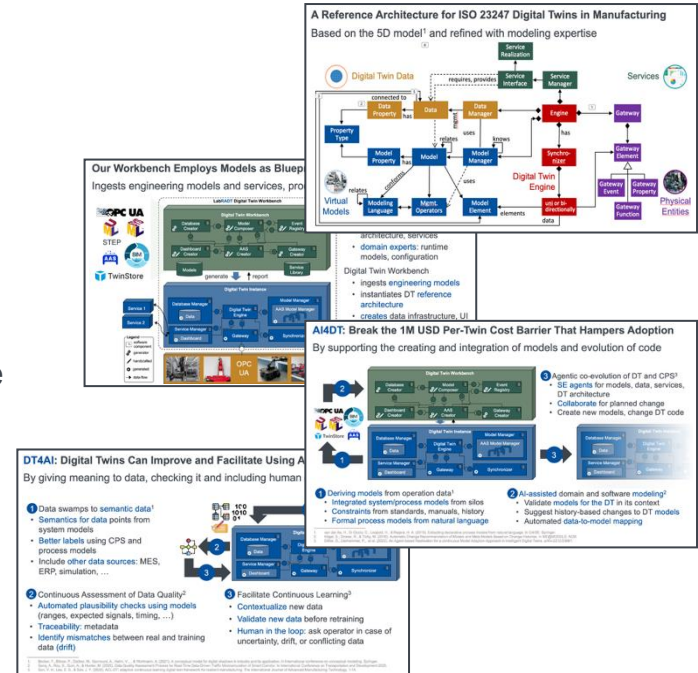


1. Kimmel, R., Michael, J., Wortmann, A., & Zhang, J. (2025). Digital Twins for Software Engineering Processes. In 47th International Conference on Software Engineering: New Ideas and Emerging Results (ICSE-NIER). IEEE.

Intelligent Digital Twins are not just 3D Models

They promise tremendous benefits and have delivered some

- **Model-driven digital twins** can help (manufacturing) SMEs
- **AI4DT**: GenAI facilitates creating model-driven digital twins
- **DT4AI**: Digital twins enable better use of AI
- Next steps for industry
 - **Semanticise**: give meaning to data via models
 - **Contractize**: DT interfaces as **subject of supplier contracts**
 - **Monetize**: twin-enabled machines, predictive maintenance subscriptions, remote commissioning, retrofit twins, ...
- Next steps for politics
 - Strategic program for **validated, interoperable, intelligent industrial DTs** (parallel to Smart Manufacturing 3.0)
 - Enforce a **reference architecture for public projects**



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